

EBU6505 – Reasoning and Agents

Block 1, Day 2 – Knowledge Representation

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Teaching Week 3

What have we covered so far this week?

- Sequential decision making and MDP components
- Return, discounting, value, and Bellman intuition
- Why reward and state matter in sequential decision problems

Main idea: Yesterday introduced sequential decision models and reward over time. Today is about storing facts in a form an agent can search and reason over.

What are we going to cover today?

- Why AI systems need explicit knowledge representation
- Entities, relations, attributes, triples, and identifiers
- Schema graphs versus instance graphs
- Why structured knowledge helps search, retrieval, and agent memory

Main idea: The goal of today is simple: take a normal sentence and turn it into a machine-usable structure.

- *Artificial Intelligence: Foundations of Computational Agents*, David Poole and Alan Mackworth (3rd ed.)
- Chapter 16 – knowledge graphs, graph modelling, and structured knowledge
- *Artificial Intelligence: A Modern Approach*, Russell and Norvig (4th ed.)
- Chapter 10 – knowledge representation

Main idea: Read the textbooks for the formal view. In class, we focus on the intuition and the patterns used in modern AI systems.

Learning Outcomes for Today

By the end of this session, you should be able to:

- explain why plain text alone is often not enough for reasoning systems
- represent simple facts as triples
- distinguish entities, relations, attributes, and identifiers
- explain how search systems and modern agents use structured knowledge

Warm-up Exercise

Question: Can you write one real-world fact so that a machine could store it clearly?

- Try one short example before the lecture continues.
- Write the fact in a way that removes ambiguity.

Try one fact now

subject



relation



object

Example:

Beijing – capitalOf – China

Main idea: A good representation is not just readable by humans. It must also be clear enough for a machine to reuse later.

Why Represent Knowledge?

Question: Why is plain language alone often not enough for an AI agent?

- Human language is rich, but it is also ambiguous.
- The same name can refer to different people or places.
- A machine must store facts in a form it can search and compare later.
- If the structure is weak, retrieval and reasoning become unreliable.

Main idea: Knowledge representation gives the machine a stable way to store facts, connect facts, and use facts again later.

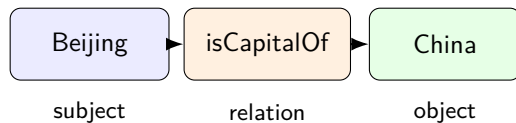
From Sentence to Structure

Question: How can one sentence become something a machine can reuse?

Human sentence

“Beijing is the capital of China.”

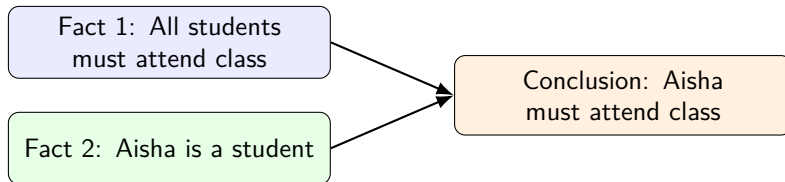
- Humans read this as one natural sentence.
- Machines do better when the parts are explicit.



Main idea: Representation breaks one sentence into clear parts so the machine can store the fact and connect it later.

Reasoning Needs Explicit Facts

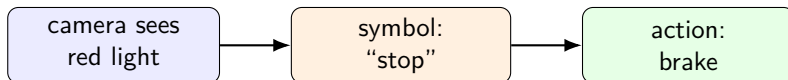
Question: How can a machine go from known facts to a new conclusion?



Main idea: If the system stores facts and rules explicitly, it can combine them step by step to derive a new answer.

Symbols Connect Perception to Action

Question: Why do symbols matter when an AI system must act in the world?



- Sensors produce raw data.
- Representation maps that raw data to a meaningful symbol.
- The symbol helps the agent choose the correct action.

Main idea: Representation is the bridge between perception and decision.

Types of Knowledge an Agent Needs

Question: Does one useful agent need only factual knowledge?

Type	Simple meaning and example
Declarative	facts about the world. Example: "Beijing is in China."
Procedural	how to do something. Example: how to book a ticket
Semantic	concepts and relations. Example: student – enrolledIn – module
Episodic	memory of past events. Example: the last support ticket
Meta-knowledge	what is known, missing, or uncertain. Example: "I need more evidence."

Main idea: Real agents mix facts, procedures, concepts, memories, and uncertainty about what they know.

Core Terms for Today

Question: What six words should you keep in mind during this lecture?

Term	Meaning
Knowledge representation	A formal way to store meaning
Entity	A thing, person, place, or object
Relation	A named link between two entities
Attribute	A property of one entity
Triple	Subject – predicate – object fact
Identifier	A unique label for one entity

Main idea: If you understand these six terms, the rest of the lecture becomes much easier to follow.

Personal Assistant Example

Question: What must an assistant understand in a query like “Play a song by Da Zhang Wei”?

User query

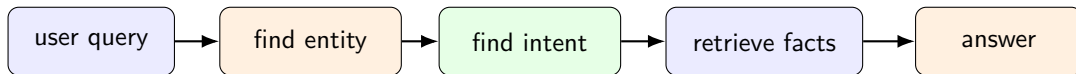
“Play a song by Da Zhang Wei”

- detect the entity: Da Zhang Wei
- detect the intent: play music
- retrieve the linked facts: artist, songs, platform, permissions

Main idea: The assistant must identify what the user means, not only what words the user typed.

Assistant Understanding Pipeline

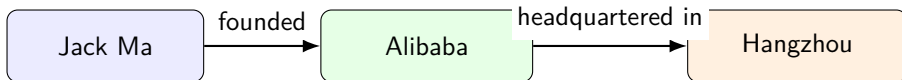
Question: What steps happen between the user's question and the system's answer?



Main idea: A good answer is usually the result of several small steps, not one magic jump from text to text.

Why Search Uses Connected Facts

Question: Why do search systems connect facts instead of storing isolated lines of text?

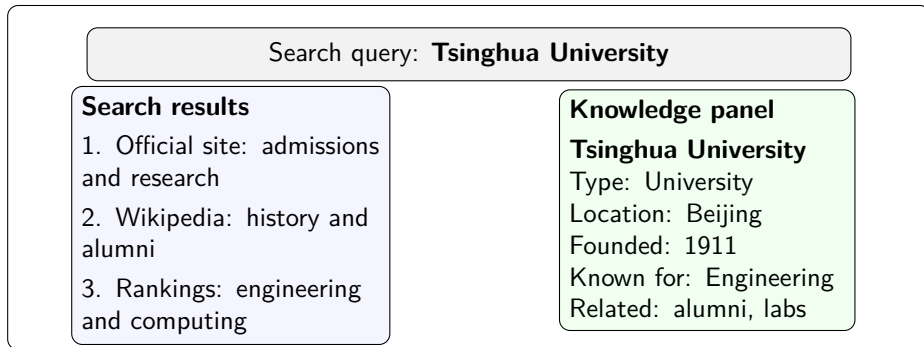


Question: "Who founded Alibaba?"
The graph already contains the path to the answer.

Main idea: Connected facts let the system move from one entity to another instead of scanning unrelated text every time.

Example: Knowledge Panel in Search

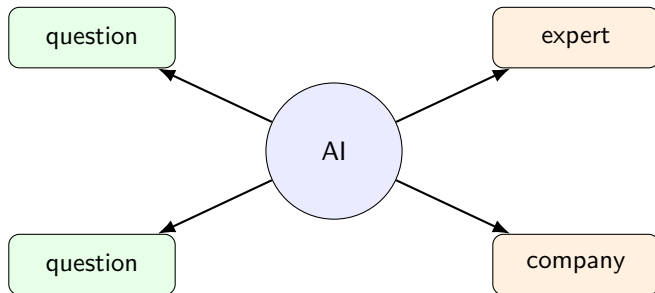
Question: What does structured knowledge look like in a real search result?



Main idea: A knowledge panel is a user-facing view of structured facts.

Topic Pages Use the Same Idea

Question: What changes when a platform links topics, questions, and people?



Main idea: Once items are linked by meaning, the system can organise a topic page instead of showing random isolated content.

Where Do We See Knowledge Representation Already?

Question: In which everyday AI systems do we already see the same representation idea?

Assistant

Maps query to
entity + intent + facts

Search

Uses linked facts
for panels and direct answers

Enterprise agent

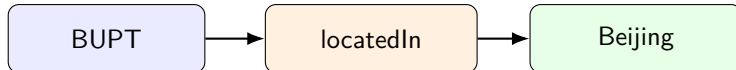
Connects documents,
people, systems, and
policies

Main idea: The products look different, but the underlying idea is the same: store meaning in a structure the system can follow.

Triple Representation

Question: What is the smallest useful fact unit in many graph systems?

(subject, predicate, object)



Symbols used on this slide

subject: the thing we are talking about

predicate: the named relation

object: the thing or value linked by that relation

Main idea: A triple is small, simple, and reusable. Many large knowledge graphs are built from many triples.

Entities, Relations, and Attributes

Question: How do we tell apart a thing, a link, and a property?

Entity

A thing in the world

Example: "Aisha"

"QMUL"

"Module EBU6505"

Relation

A link between entities

Example: enrolledIn

teaches

locatedIn

Attribute

A property of one entity

Example: birthDate

moduleCode

roomNumber

Main idea: Do not mix them up: entity = thing, relation = link, attribute = property.

Why Identifiers Matter

Question: Why is a unique identifier better than only a name string?

Problem

Two different people can both be called “Ali”. The name is helpful for humans, but it is not enough for the machine.

Ali
ID: person_104

Ali
ID: student_022

- names are human-friendly
- identifiers keep different entities separate

Main idea: Names help users. Identifiers help the machine keep different entities separate.

Schema Graph Versus Instance Graph

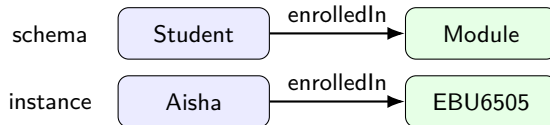
Question: What is the difference between the pattern of a graph and the actual facts inside it?

Schema graph

Defines the allowed kinds of nodes and links.

Instance graph

Stores one real fact that follows that pattern.



Main idea: Schema gives the pattern. Instance data fills that pattern with real facts.

From Knowledge Representation to Agent Memory

Question: Why do modern agents need structured memory instead of only prompt text?

Prompt-only memory

Long, noisy, expensive to search
Easy to forget exact facts

Structured memory

Facts are normalized and linked
Easier to retrieve exact evidence

Main idea: A long prompt is temporary context. A structured memory is reusable knowledge.

Memory Layers in Modern Agents

Question: What kinds of memory should a useful agent keep?

working memory – current task context

episodic memory – past events and outcomes

semantic memory – normalized facts and relations

tool memory – what tools exist and when to use them

Main idea: Not all memory is the same. Some memory is short-term, while some memory must survive across many tasks.

Why Structured Memory Beats Raw Logs

Question: Why is a graph or table often better than a long chat log?

Raw log

- Repeated wording
- Missing exact links
- Hard to audit

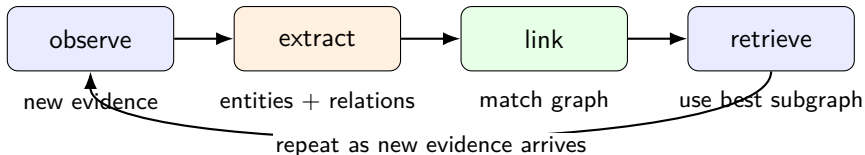
Structured memory

- Explicit entities and links
- Easier retrieval
- Clearer provenance

Main idea: Structured memory reduces ambiguity and makes later reasoning easier to inspect.

Agent Memory Update Loop

Question: How does an agent turn new evidence into usable memory?

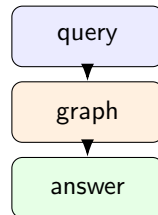


Main idea: Useful memory is not static. The agent keeps updating the graph as it learns new evidence.

Case Study: Search Engines Reason Over KGs

Question: How can a search engine use a graph to answer faster and more clearly?

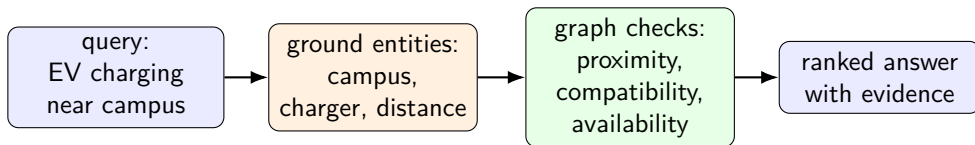
- map the query to entities
- follow useful relation paths
- rank candidate answers
- show a short answer or panel



Main idea: The graph helps the system move quickly from a query to a supported answer path.

Search Example Workflow

Question: How might a graph help with “best EV charging options near campus”?



Main idea: The graph lets the search system combine several conditions instead of matching only keywords.

Search Engine KG Failure Modes

Question: What can go wrong if the stored knowledge is weak or outdated?

Common failures

- stale facts
- poor entity linking
- missing facts
- missing provenance

Common fixes

- freshness checks
- confidence thresholds
- source traces
- human review for risky cases

Main idea: A knowledge graph is useful only if the links are accurate, current, and traceable.

Agents Reasoning Over Internal Company Knowledge

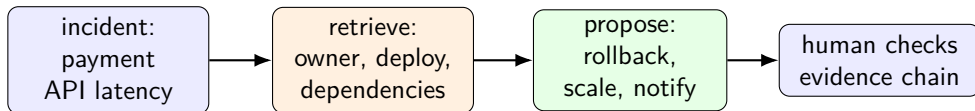
Question: Why do companies want agents to reason over internal knowledge?

- useful facts are spread across documents, tickets, systems, and people
- the agent must connect services, owners, changes, and policies
- this helps answer questions like “who owns service X?” or “what changed last week?”

Main idea: Enterprise knowledge is not one document. It is a network of connected facts spread across many systems.

Enterprise Reasoning Example

Question: What might an internal support agent do during an incident?



Main idea: The agent does not act blindly. It gathers linked evidence first, then proposes a grounded next step.

Governance for Internal Knowledge Agents

Question: Why can an internal knowledge agent not be given unlimited freedom?

- role-based access control
- provenance for each retrieved fact
- privacy and security filters
- human review for high-risk actions

Main idea: Internal knowledge is powerful, so the agent must be accurate, traceable, and controlled.

Modern AI Assistants Reason About Tool Choice

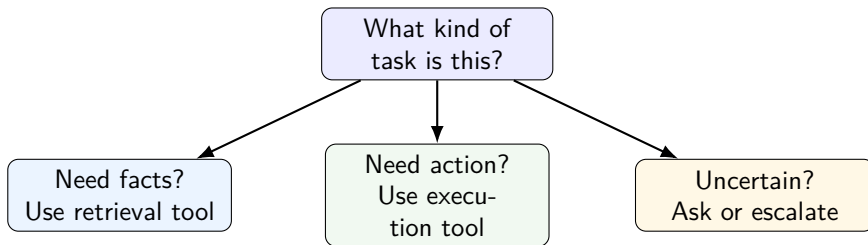
Question: Before calling a tool, what does an assistant need to decide?

- do I need a factual lookup?
- do I need to perform an action?
- am I uncertain enough that I should ask a clarifying question first?

Main idea: Good agents do not call tools randomly. They first decide what kind of job they are trying to do.

Tool Selection Policy Pattern

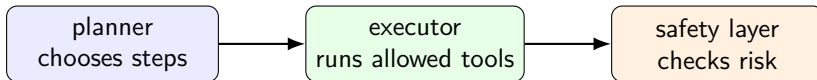
Question: What simple policy can an agent follow when choosing a tool?



Main idea: A simple decision policy already prevents many bad tool calls.

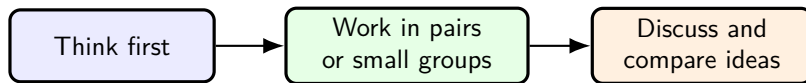
Why Separate Planning From Execution?

Question: Why do many agent systems keep planning and execution separate?



Main idea: This separation makes the system easier to control, debug, and audit.

In-Class Exercises



Use this time to work on today's in-class tasks and prepare one clear answer you can discuss.

Main idea: Keep the graph small. A clear tiny graph is better than a large messy one.

Key Takeaways

Question: What should you remember most from today's lecture?

- AI systems need explicit knowledge structures, not only natural language.
- Triples are a simple and powerful way to store facts.
- Entities, relations, attributes, and identifiers must be distinguished clearly.
- Search systems and modern agents both benefit from structured knowledge.

Main idea: If a machine can store facts clearly, it can retrieve better and reason better.

Further Study

- Read Poole Chapter 16 and AIMA Chapter 10.
- Practice: Poole Ch.16, Exercises 16.2, 16.3, 16.5, 16.6, and 16.9.
- AIMA knowledge-representation exercise page: 2, 9, 13, 16, and 26.
- Try writing 5 triples for one domain you know well.
- Ask yourself: which facts are entities, which are attributes, and which are relations?
- Prepare for Day 3: we move from simple representation to stronger graph foundations.